

Introduction to LISP

- LISP stands for "List processing". It is the second-oldest high-level programming language.
- It is invented by John McCarthy in the year 1958 at the Massachusetts Institute of Technology.
- It is particularly suitable for Artificial Intelligence programs, as it processes symbolic information effectively.
- It serves as a common language, which can be easily extended for specific implementation.
- Programs written in common LISP do not depend on machine-specific characteristics.

Features of Common LISP

- It is machine-independent.
- It uses iterative design methodology, and easy extensibility.
- It allows updating the programs dynamically.
- It provides high level debugging.
- It provides advanced object-oriented programming.
- It provides a convenient macro system.
- It provides wide-ranging data types like objects, structures, lists, vectors, hash-tables and symbols etc.
- It is expression-based.
- It provides extensive control structure.

LISP- Basic Syntax

LISP programs are made up of three basic building blocks.

Atom: An atom includes numbers, special characters, and strings.

e.g.: hello-from-tutorials point,
name, 123008907, 'hello',
Block #221, abc 123,

List: A list is a sequence of numbers or string contiguous characters in parentheses.

e.g.: (I am a list), (a(abc)d
e fgh).

String: A string is a group of characters enclosed in double quotation marks.

e.g.: "I am a string", a b a c d
e f g #8020181"

Use of Single Quotation Mark

Using ~~a~~ single quote:

A single quote is used in representing data literal. It returns data.

(write('2 3')) Result = ('2 3')

Without using a single quote,

Whatever it is, so then the function is applied to the value of the variable.

(write('2 3')) Result = 6

LISP Functions

A function is a group of statements that together perform a task.

Defining functions in LISP.

The macro named defun is used for defining functions.

The defun macro needs three arguments.

Name of the function.

Parameters of the function.

Body of the function.

(write (averagenum 10 20 30 40))

You can provide an empty list as parameters, which means the function takes no arguments, the list is empty, written as ().

LISP also allows optional, multiple, and keyword arguments.

The documentation string describes the purpose of the function. It is associated with the name of the function and can be obtained using the documentation function.

The body of the function may consist of any number of Lisp expressions.

The value of the last expression in the body is returned as the value of the function.

You can also return a value from the function using the return-from special operator.

LISP-Comparison Operators

LISP comparison operators may take more than two operands.

These operators are used to compare numbers.

The comparison symbols are (=, <, >, <=, >=, /=, min, max).

LISP-Comparison Functions

These functions returns either True (T) or Nil (false)

$(4=10) \rightarrow \text{NIL}$

$(4 \neq 10) \rightarrow \text{T}$

$(4 > 10) \rightarrow \text{NIL}$

$(4 < 10) \rightarrow \text{T}$

$(4 \geq 10) \rightarrow \text{NIL}$

$(4 \leq 10) \rightarrow \text{T}$

Other Comparison Functions

Max: This operator returns the maximum value

$(\text{max } 10 \ 15 \ 20 \ 180) \rightarrow 180$

Min: This operator returns the minimum value

$(\text{min } 5 \ 15 \ 18 \ 8 \ 9) \rightarrow 3$

DATE: _____

DAY: _____

Assigning Functions

Syntax: setq (argument)

Ex (setq a) →

(°a 4) → 12

Explicit evaluation call;

(eval a) → 3

Applications Built on LISP

Large successful applications
built in Lisp

Emacs ✓

G2 ✓

AutoCad ✓

Igor Engraver ✓

Yahoo store ✓